Quiz — 2.1.3 Thinking Procedurally

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 · Component 02 · 2.1.3

Section A — Multiple choice (6 marks)

1. Breaking the task "build a shopping website" into "user accounts", "product catalogue", "basket" and "checkout" sub-problems is an example of: A. Abstraction B. Decomposition C. Caching D. A race condition Your answer: ☐

2. In a "make a cup of tea" procedure, which step *must* come before "pour water onto the teabag"? A. Add milk B. Boil the water C. Drink the tea D. Wash the cup afterwards Your answer: ☐

3. A self-contained block of code that performs one named task (e.g. `calculateVAT`) and can be called from several places is best described as a: A. Precondition B. Sub-procedure C. Cache D. Decision point Your answer: ☐

4. A robot must "navigate to the box" before it can "pick up the box". This dependency means the two steps must be: A. Run concurrently B. Run in a fixed order C. Abstracted away D. Cached Your answer: ☐

5. Why is decomposition useful when several programmers build one large system together? A. It guarantees the program runs on multiple cores B. Each sub-problem can be assigned to and worked on by a different person C. It removes all the inputs and outputs D. It makes data automatically sorted Your answer: ☐

6. In a payroll program the order is: read hours → calculate gross pay → deduct tax → output payslip. "Deduct tax" must follow "calculate gross pay" because: A. The two steps are independent B. Tax is calculated from the gross pay produced by the earlier step C. Tax can be cached D. The payslip is an input Your answer: ☐

Section B — Short answer (9 marks)

7. A developer is building a vending machine program. Identify **three** sensible sub-procedures the problem could be decomposed into. **[3]**

8. For an ATM "withdraw cash" task, state **two** steps that must happen in a fixed order and explain why one depends on the output of the other. **[2]**

9. Explain **two** advantages of dividing a large program into sub-procedures rather than writing it as one long block of code. **[2]**

10. A quiz game program calls the same `shuffleQuestions` sub-procedure at the start of every round. Explain why writing this as a sub-procedure is better than copying the shuffling code into each round. **[2]**

Section C — Applied question (5 marks)

11. A school wants a program that produces end-of-term reports for every pupil. Using *thinking procedurally*, decompose this task into named sub-procedures and put them in the correct order. For at least one pair of steps, explain why the order is fixed (i.e. one step needs the output of an earlier step). **[5]**